

What the Theory Never Specifies

Preregistered Simulations of REBUS and ALBUS

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The question, the agent, the rules

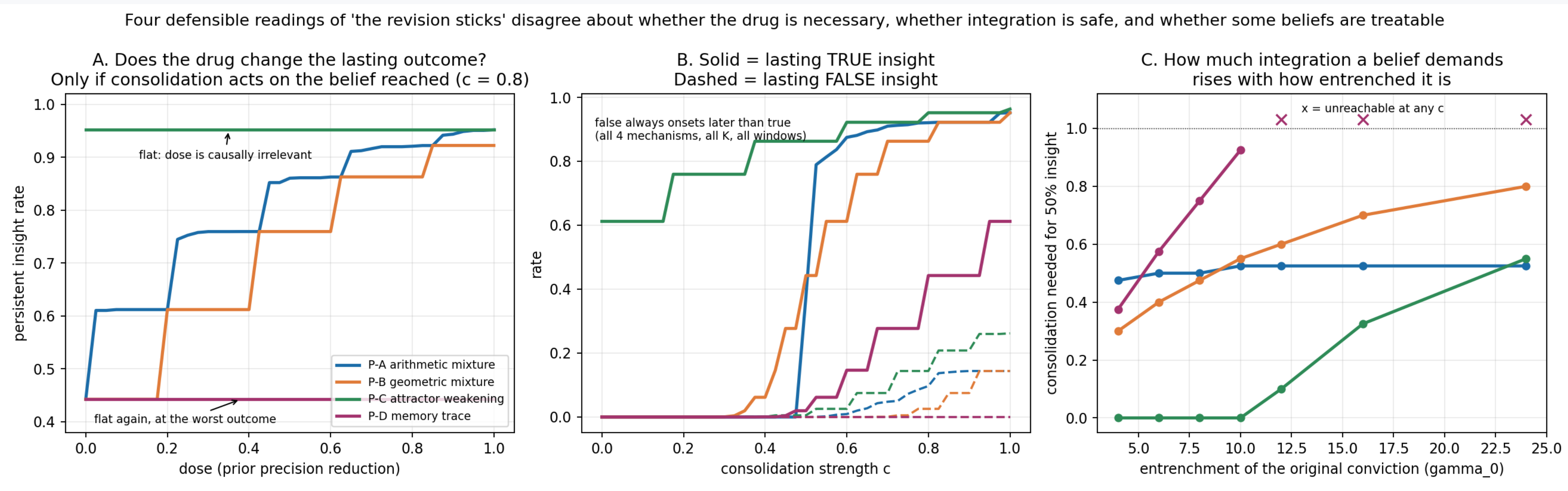
Why do some psychedelic sessions produce lasting change while others change nothing, or leave a person more convinced of the belief that harms them? REBUS says the drug relaxes rigid priors so beliefs can revise. ALBUS says low doses strengthen them. Neither theory has ever been implemented.

The agent is the minimal Bayesian belief-updater both theories describe: six hypotheses, one true, one a maladaptive conviction held at 3000 to 1 odds, and a dose that does exactly one thing, relax the prior's weight while evidence arrives: $\gamma(d) = \gamma_0(1 - d)$.

The rules: every mapping was committed to a public git history before the code that tested it existed, across 11 amendments and 44 numbered predictions. Predictions that failed are reported on this poster, because the failures produced the best results.

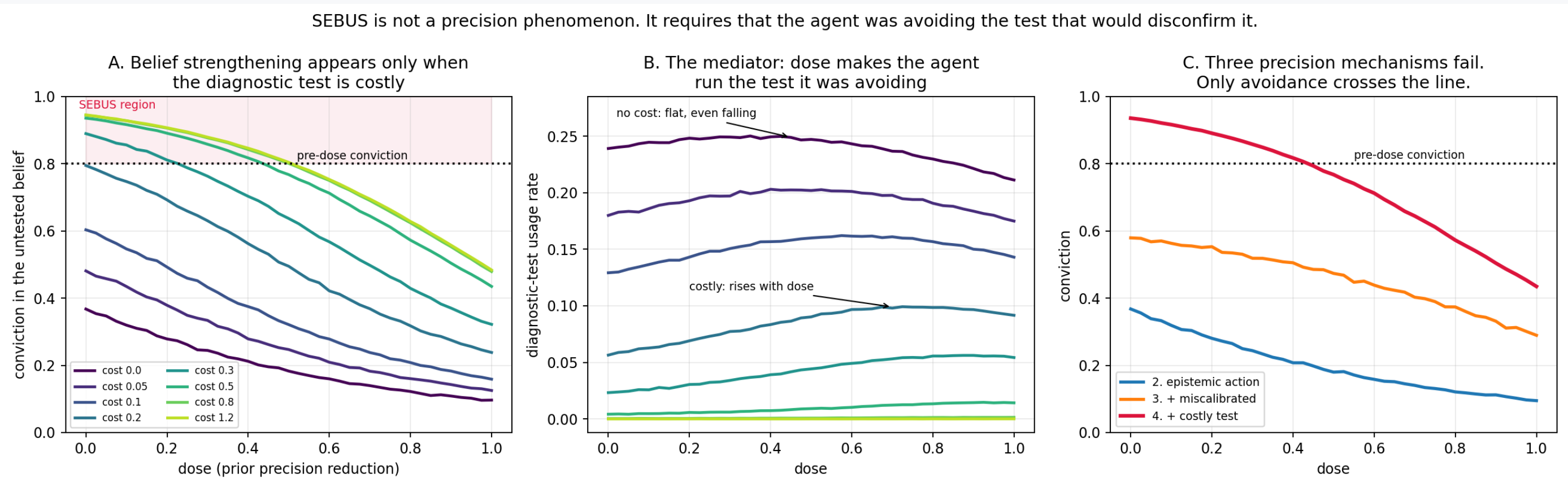
Scale: roughly three billion simulated agents across eleven arms, all local compute, zero API spend.

Nothing lasts as stated, and an unstated assumption decides whether the drug matters



Literal REBUS produces zero lasting change: dose enters only through a precision that reverts, so it cannot reach the outcome. Consolidation must be added, and it acts as a gate. But four defensible readings of what consolidation operates on disagree about whether the drug matters at all (under the attractor reading, dose zero is best), whether integration can be made safe, and whether entrenched beliefs are treatable. Neither REBUS nor REBAS states which reading is right; each is measurable.

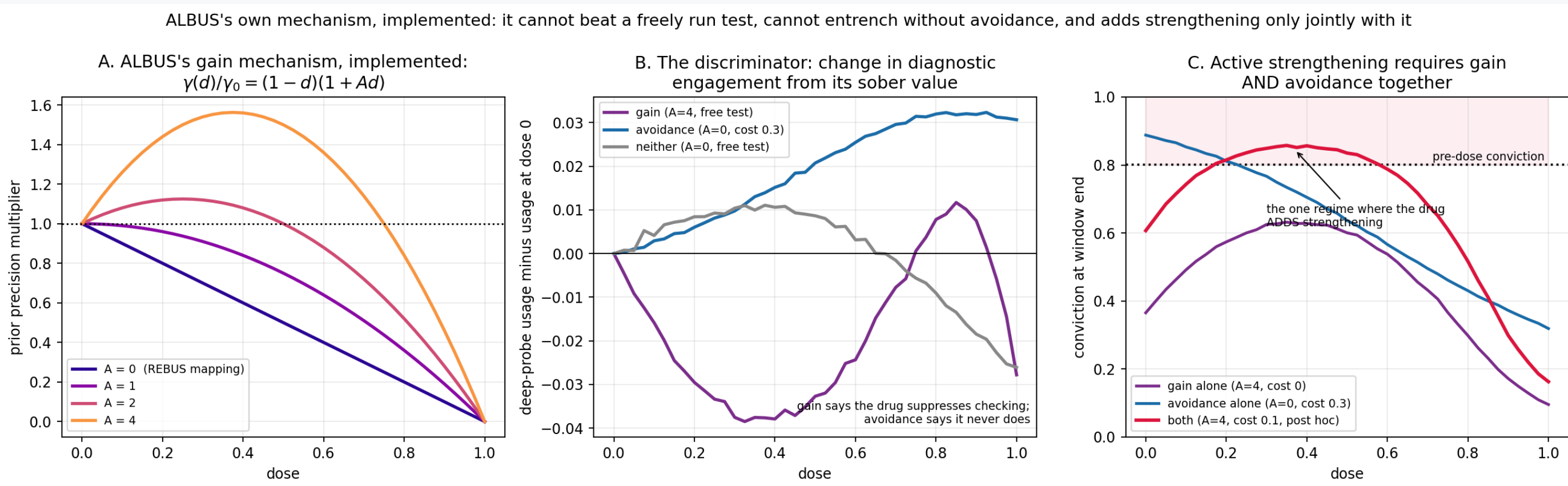
Strengthening is avoidance



SEBUS never emerges from precision mechanics: passive relaxation, epistemic action, and miscalibrated self-knowledge all fail. It appears only when the one diagnostic test carries a cost the agent declines to pay. The mediator shows the drug's derivable role: relaxation raises the value of the avoided test until it is worth running. Above a cost ceiling, no dose helps.

The numbers: SEBUS appears between cost 0.2 and 0.3; at cost 1.2 the test never runs and no dose helps.

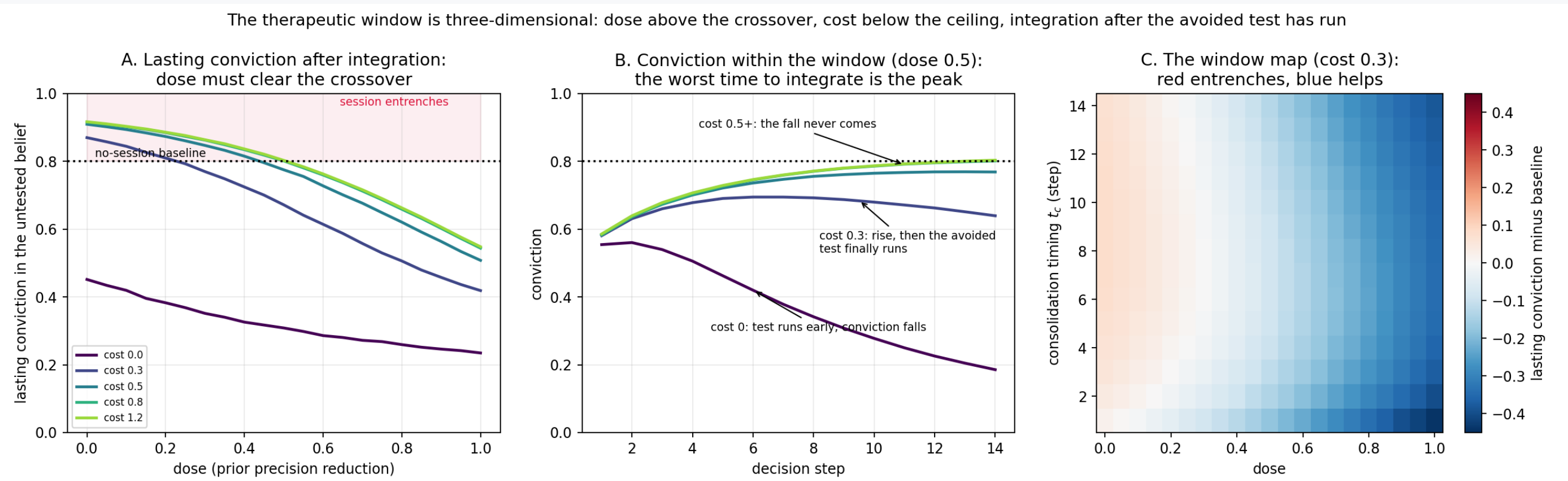
ALBUS's own mechanism needs avoidance too



A low-dose gain increase, implemented honestly, cannot beat a freely run test and cannot entrench without a cost. The accounts separate by sign: gain says the drug suppresses checking at low dose, avoidance says it never does. Drug-added strengthening exists in one confirmed regime only: gain and cost together.

The numbers: the confirmed interior window spans doses 0.17 to 0.55 at strong gain with cost 0.1.

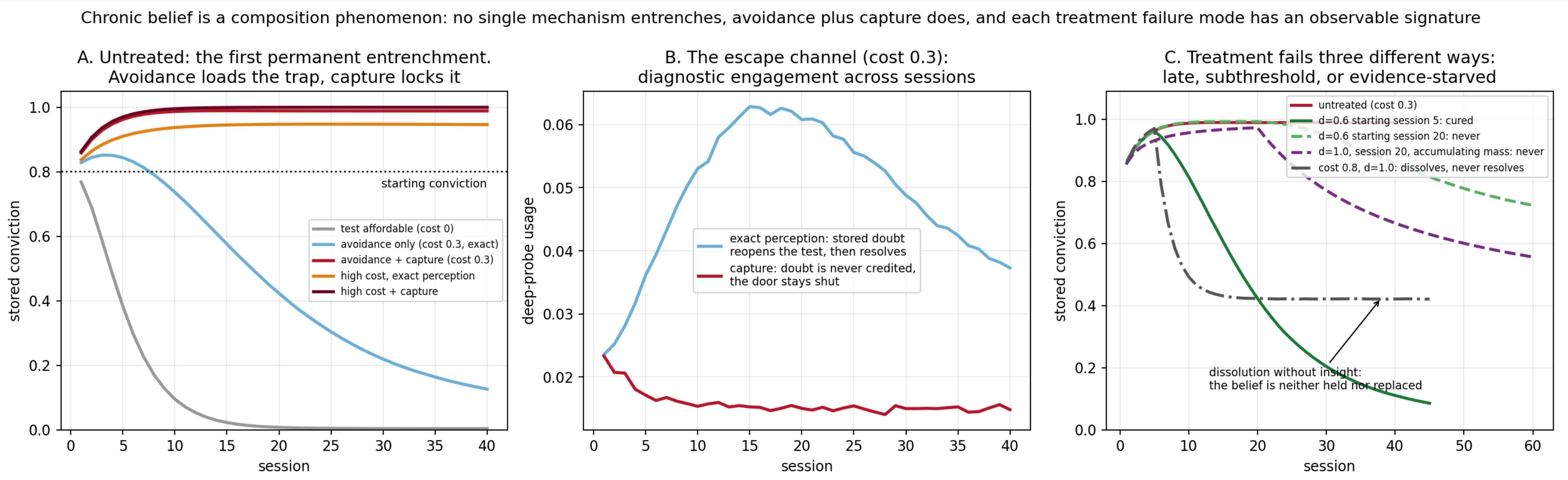
The therapeutic window is three-dimensional



Dose must clear a cost-dependent crossover, cost must sit below the ceiling, and integration must capture the state after the avoided test has run. Miss a coordinate and the same session entrenches the belief, or dissolves it without replacing it: lasting conviction falls while insight never arrives.

The numbers: lasting crossovers sit at the acute ones, doses 0.25, 0.45, 0.55 at costs 0.3, 0.5, 0.8.

The ratchet: how a belief becomes chronic



Avoidance stores up the very doubt that eventually reopens the test; capture never credits it, so conviction climbs to 0.99 and locks. **Avoidance loads the trap, capture locks it.** Treatment fails three observable ways: subthreshold, evidence-starved, or too late.

What to take home

- Integration, not dose, gates lasting change; which quantity it operates on is an unstated assumption that flips the answer.
- The drug's derivable role is to make an avoided disconfirming test worth running: the exposure account of therapy, emerging from REBUS's own vocabulary.
- Rigid beliefs defend themselves through what you do (avoidance), what you see (capture), and what counts (gain); only compositions entrench.
- One behavioral marker separates every account: engagement with avoided disconfirming evidence, by sign and by trajectory.

Belief dynamics only: no claim about phenomenal experience. The same implement-and-see discipline on machine implementations of global workspace theory gave the same lesson: the unstated assumptions decide.